

TudeDude Assignment 1

1. Creating and Renaming Files/Directories

Create a directory named `test_dir` using `mkdir`.

Inside `test_dir`, create an empty file called `example.txt`.

Rename `example.txt` to `renamed_example.txt` using `mv`

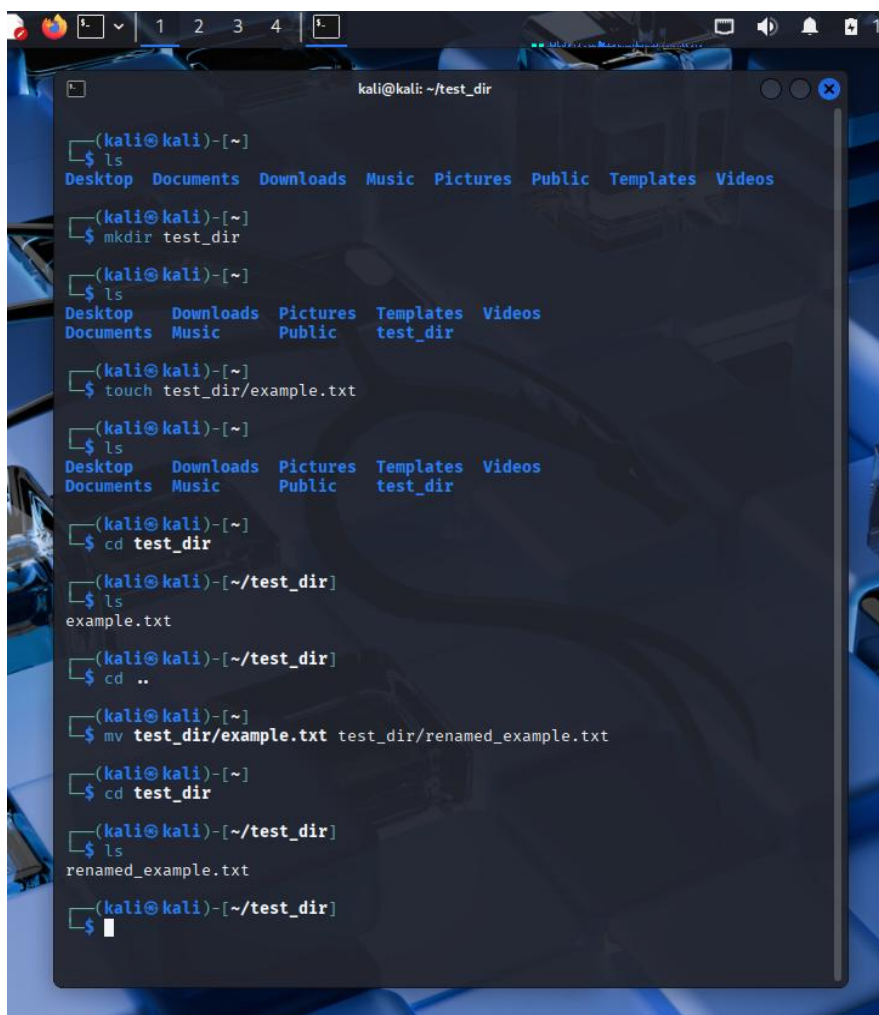
Solution:

Create directory

```
mkdir test_dir
```

```
touch test_dir/example.txt
```

```
mv test_dir/example.txt test_dir/renamed_example.txt
```



```
kali@kali: ~/test_dir

(kali@kali)-[~]
$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos

(kali@kali)-[~]
$ mkdir test_dir

(kali@kali)-[~]
$ ls
Desktop  Downloads  Pictures  Templates  Videos
Documents  Music  Public  test_dir

(kali@kali)-[~]
$ touch test_dir/example.txt

(kali@kali)-[~]
$ ls
Desktop  Downloads  Pictures  Templates  Videos
Documents  Music  Public  test_dir

(kali@kali)-[~]
$ cd test_dir

(kali@kali)-[~/test_dir]
$ ls
example.txt

(kali@kali)-[~/test_dir]
$ cd ..

(kali@kali)-[~]
$ mv test_dir/example.txt test_dir/renamed_example.txt

(kali@kali)-[~]
$ cd test_dir

(kali@kali)-[~/test_dir]
$ ls
renamed_example.txt

(kali@kali)-[~/test_dir]
$
```

2. Viewing File Contents

Use cat to display the contents of /etc/passwd.

Display only the first 5 lines of /etc/passwd using head.

Display only the last 5 lines of /etc/passwd using tail.

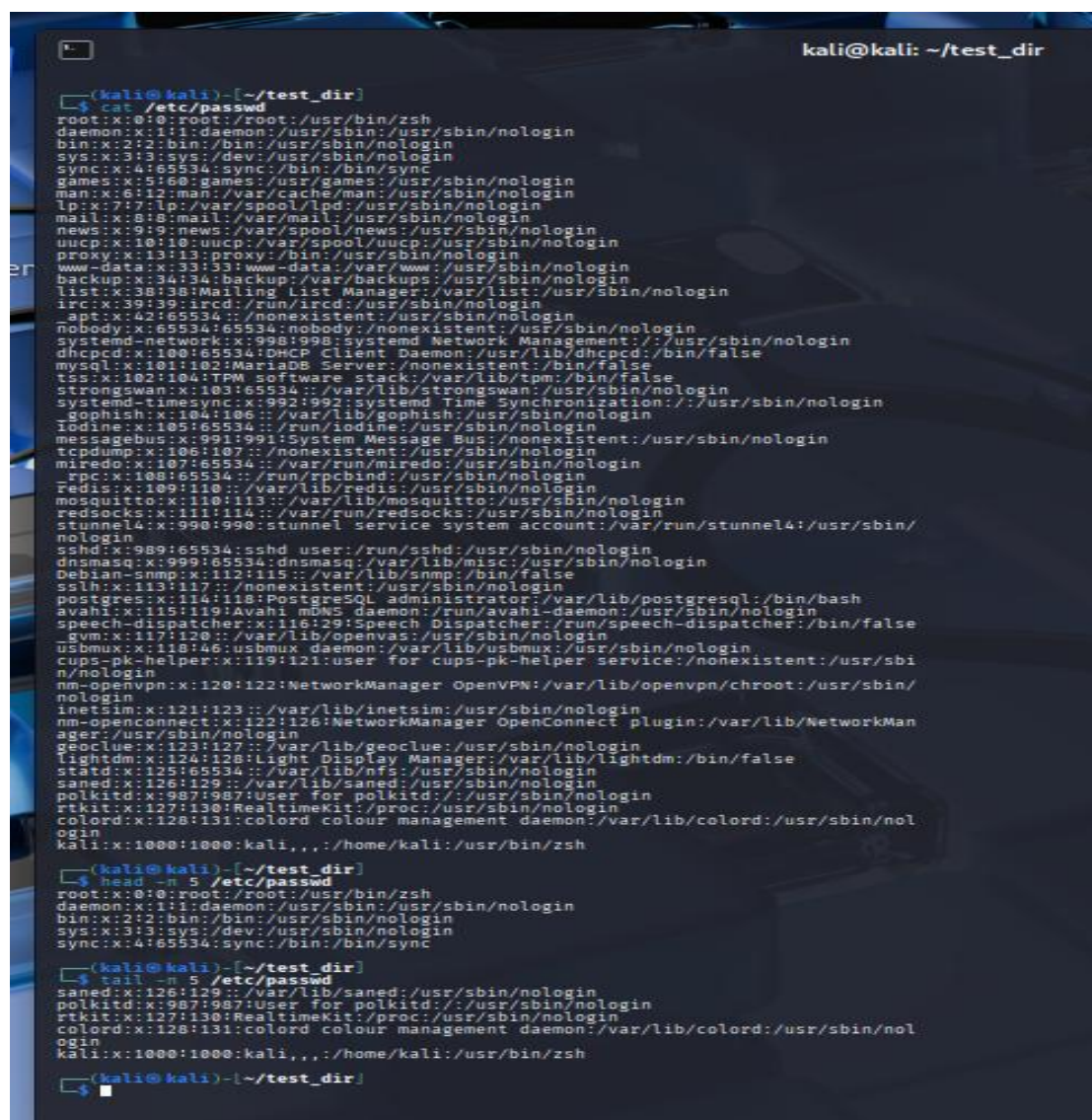
Solution:

cat /etc/passwd

head -n 5 /etc/passwd

tail -n 5 /etc/passwd

cat prints the file to stdout. head -n 5 prints the first 5 lines. tail -n 5 prints the last 5 lines.



```
kali@kali: ~/test_dir

(kali@kali)~[~/test_dir]
$ cat /etc/passwd
root:x:0:0:root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail list Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpd:/bin/false
mysql:x:101:102:MySQL Server:/nonexistent:/bin/false
tss:x:102:104:TPM software stack:/var/lib/tpm:/bin/false
strongswan:x:103:65534::/var/lib/strongswan:/usr/sbin/nologin
systemd-timesync:x:992:992:systemd Time Synchronization:/:/usr/sbin/nologin
gophish:x:104:106::/var/lib/gophish:/usr/sbin/nologin
Iodine:x:105:65534::/run/iodine:/usr/sbin/nologin
messagebus:x:991:991:system Message Bus:/nonexistent:/usr/sbin/nologin
tcpdump:x:106:107::/var/lib/tcpdump:/usr/sbin/nologin
miredo:x:107:65534::/var/run/miredo:/usr/sbin/nologin
rpc:x:108:65534::/run/rpcbind:/usr/sbin/nologin
redis:x:109:110::/var/lib/redis:/usr/sbin/nologin
mosquitto:x:110:113::/var/lib/mosquitto:/usr/sbin/nologin
redsocks:x:111:114::/var/run/redsocks:/usr/sbin/nologin
stunnel4:x:990:990:stunnel service system account:/var/run/stunnel4:/usr/sbin/nologin
sshd:x:989:65534:sshd user:/run/sshd:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/dnsmasq:/usr/sbin/nologin
Debian-snmpp:x:112:115::/var/lib/snmpp:/bin/false
ssllh:x:113:117::/nonexistent:/usr/sbin/nologin
postgres:x:114:118:PostgreSQL administrator:/var/lib/postgresql:/bin/bash
avahi:x:115:119:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
speech-dispatcher:x:116:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false
gvm:x:117:120::/var/lib/openvas:/usr/sbin/nologin
usbmux:x:118:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
cups-pk-helper:x:119:121:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin
nm-openvpn:x:120:122:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin
inetsim:x:121:123::/var/lib/inetsim:/usr/sbin/nologin
nm-openconnect:x:122:126:NetworkManager OpenConnect plugin:/var/lib/NetworkManager:/usr/sbin/nologin
geoclue:x:123:127::/var/lib/geoclue:/usr/sbin/nologin
lightdm:x:124:128:Light Display Manager:/var/lib/lightdm:/bin/false
statd:x:125:65534::/var/lib/nfs:/usr/sbin/nologin
sane:x:126:129::/var/lib/sane:/usr/sbin/nologin
polkitd:x:987:987:user for polkitd:/:/usr/sbin/nologin
rtkit:x:127:130:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:128:131:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
kali:x:1000:1000:kali,,,:/home/kali:/usr/bin/zsh

(kali@kali)~[~/test_dir]
$ head -n 5 /etc/passwd
root:x:0:0:root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync

(kali@kali)~[~/test_dir]
$ tail -n 5 /etc/passwd
sane:x:126:129::/var/lib/sane:/usr/sbin/nologin
polkitd:x:987:987:user for polkitd:/:/usr/sbin/nologin
rtkit:x:127:130:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:128:131:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
kali:x:1000:1000:kali,,,:/home/kali:/usr/bin/zsh

(kali@kali)~[~/test_dir]
$
```

3. Searching for Patterns

Use **grep** to find all lines containing the word "root" in **/etc/passwd**.

Solution: **grep 'root' /etc/passwd**

grep searches for a pattern and prints matching lines. Quotes around **root** are optional here but recommended.

A terminal window titled 'kali@kali: ~/test_dir' showing the command 'grep 'root' /etc/passwd' being executed. The output displays two lines from the /etc/passwd file: 'root:x:0:0:root:/root:/usr/bin/zsh' and 'nm-openvpn:x:120:122:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin'.

```
kali@kali: ~/test_dir
(kali@kali)-[~/test_dir]
$ grep 'root' /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
nm-openvpn:x:120:122:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin
(kali@kali)-[~/test_dir]
$
```

4. Zipping and Unzipping

Compress the **test_dir** directory into a file named **test_dir.zip** using **zip**.

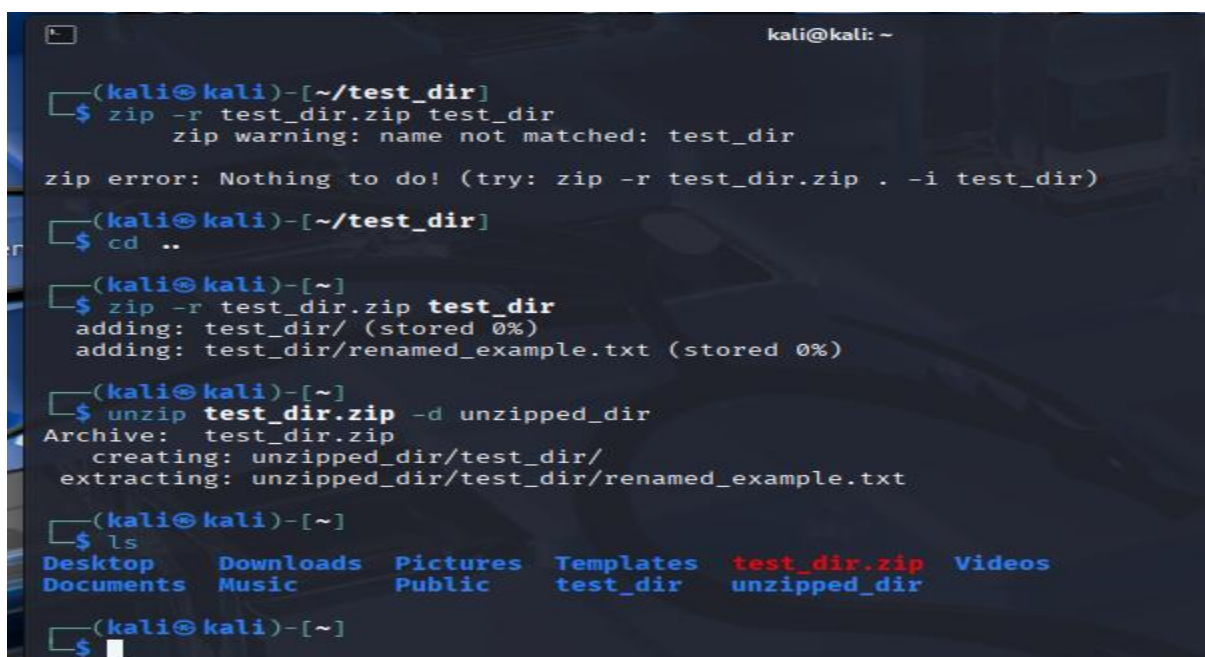
Unzip **test_dir.zip** into a new directory named **unzipped_dir**.

Solution:

zip -r test_dir.zip test_dir

unzip test_dir.zip -d unzipped_dir

zip -r packages directory contents recursively into **test_dir.zip**. **unzip -d <dir>** extracts into the named directory.

A terminal window titled 'kali@kali: ~' showing the process of creating and extracting a zip file. First, 'zip -r test_dir.zip test_dir' is run, resulting in a warning and an error. Then, 'cd ..' is used to move to the parent directory. Next, 'zip -r test_dir.zip test_dir' is run again, successfully creating the zip file. Finally, 'unzip test_dir.zip -d unzipped_dir' is run, successfully extracting the contents into the 'unzipped_dir' directory. A 'ls' command at the end shows the files in the current directory.

```
kali@kali: ~
(kali@kali)-[~/test_dir]
$ zip -r test_dir.zip test_dir
zip warning: name not matched: test_dir
zip error: Nothing to do! (try: zip -r test_dir.zip . -i test_dir)
(kali@kali)-[~/test_dir]
$ cd ..
(kali@kali)-[~]
$ zip -r test_dir.zip test_dir
adding: test_dir/ (stored 0%)
adding: test_dir/renamed_example.txt (stored 0%)
(kali@kali)-[~]
$ unzip test_dir.zip -d unzipped_dir
Archive: test_dir.zip
creating: unzipped_dir/test_dir/
extracting: unzipped_dir/test_dir/renamed_example.txt
(kali@kali)-[~]
$ ls
Desktop    Downloads  Pictures   Templates  test_dir.zip  Videos
Documents  Music      Public     test_dir   unzipped_dir
(kali@kali)-[~]
$
```

5. Downloading Files

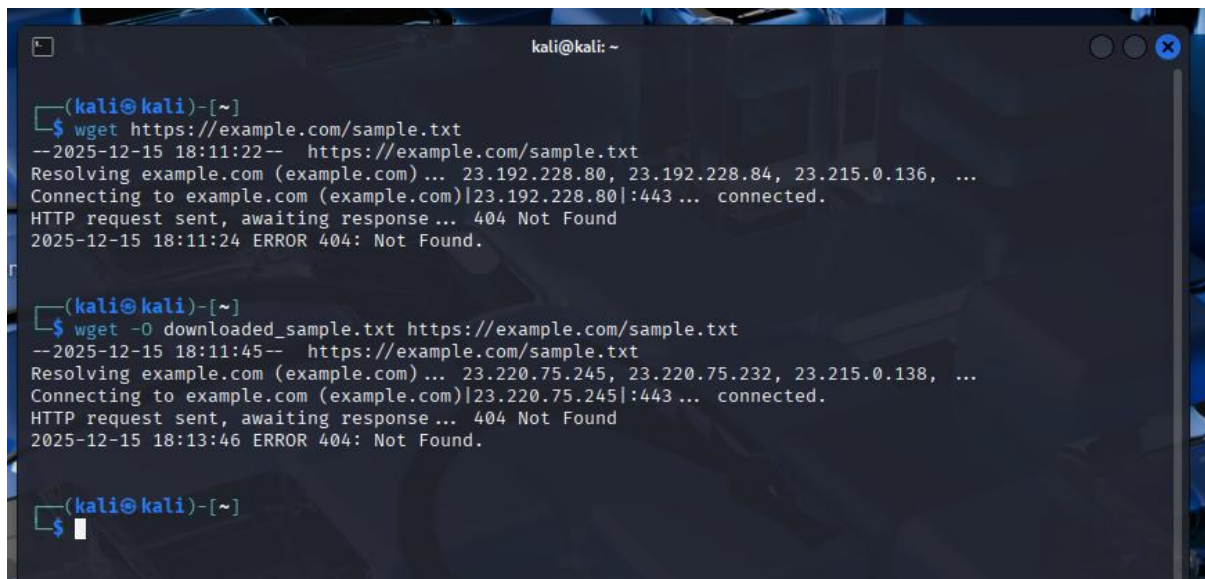
Use `wget` to download a file from a URL (e.g., <https://example.com/sample.txt>).

Solution:

`wget https://example.com/sample.txt`

`wget -O downloaded_sample.txt https://example.com/sample.txt`

`wget` URL downloads a file. `-O <name>` writes to a chosen filename.

A terminal window titled 'kali@kali: ~' showing two failed wget attempts. The first attempt uses 'wget https://example.com/sample.txt' and the second uses 'wget -O downloaded_sample.txt https://example.com/sample.txt'. Both result in '404 Not Found' errors. The terminal output for the first command is: (kali@kali)-[~]
\$ wget https://example.com/sample.txt
--2025-12-15 18:11:22-- https://example.com/sample.txt
Resolving example.com (example.com) ... 23.192.228.80, 23.192.228.84, 23.215.0.136, ...
Connecting to example.com (example.com)|23.192.228.80|:443 ... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-12-15 18:11:24 ERROR 404: Not Found.
The output for the second command is: (kali@kali)-[~]
\$ wget -O downloaded_sample.txt https://example.com/sample.txt
--2025-12-15 18:11:45-- https://example.com/sample.txt
Resolving example.com (example.com) ... 23.220.75.245, 23.220.75.232, 23.215.0.138, ...
Connecting to example.com (example.com)|23.220.75.245|:443 ... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-12-15 18:13:46 ERROR 404: Not Found.
The prompt is now (kali@kali)-[~]
\$

6. Changing Permissions

Create a file named `secure.txt` and change its permissions to read-only for everyone using `chmod`.

Solution:

`touch secure.txt`

`ls -l`

`chmod 444 secure.txt`

`ls -l secure.txt`

`chmod 444` sets read permission for owner/group/others, no write or execute. `ls -l` shows permissions.


```
kali@kali: ~  
  
(kali@kali)-[~]  
$ ls  
Desktop      downloaded_sample.txt  Music      Public      test_dir      unzipped_dir  
Documents    Downloads              Pictures    Templates    test_dir.zip  Videos  
  
(kali@kali)-[~]  
$ touch secure.txt  
  
(kali@kali)-[~]  
$ ls -l  
total 44  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Desktop  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Documents  
-rw-rw-r-- 1 kali kali   0 Dec 15 18:11 downloaded_sample.txt  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Downloads  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Music  
drwxr-xr-x 2 kali kali 4096 Dec  4 23:04 Pictures  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Public  
-rw-rw-r-- 1 kali kali   0 Dec 15 18:17 secure.txt  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Templates  
drwxrwxr-x 2 kali kali 4096 Dec 15 17:44 test_dir  
-rw-rw-r-- 1 kali kali 352 Dec 15 18:00 test_dir.zip  
drwxrwxr-x 3 kali kali 4096 Dec 15 18:00 unzipped_dir  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Videos  
  
(kali@kali)-[~]  
$ chmod 444 secure.txt  
  
(kali@kali)-[~]  
$ ls -l  
total 44  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Desktop  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Documents  
-rw-rw-r-- 1 kali kali   0 Dec 15 18:11 downloaded_sample.txt  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Downloads  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Music  
drwxr-xr-x 2 kali kali 4096 Dec  4 23:04 Pictures  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Public  
-r--r--r-- 1 kali kali   0 Dec 15 18:17 secure.txt  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Templates  
drwxrwxr-x 2 kali kali 4096 Dec 15 17:44 test_dir  
-rw-rw-r-- 1 kali kali 352 Dec 15 18:00 test_dir.zip  
drwxrwxr-x 3 kali kali 4096 Dec 15 18:00 unzipped_dir  
drwxr-xr-x 2 kali kali 4096 Oct  8 00:35 Videos  
  
(kali@kali)-[~]  
$ ls -l secure.txt  
-r--r--r-- 1 kali kali 0 Dec 15 18:17 secure.txt  
  
(kali@kali)-[~]  
$
```

7. Working with Environment Variables

Use export to set a new environment variable called MY_VAR with the value "Hello, Linux!".

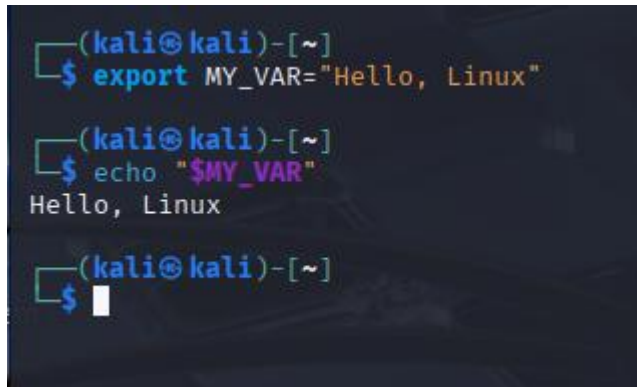
Solution:

```
export MY_VAR="Hello, Linux!"
```

Verify it is set

echo "\$MY_VAR"

export VAR="value" sets the variable for the current shell session and any child processes. echo prints it.

A terminal window with a dark background and light blue text. The prompt is (kali@kali)-[~]. The first command is \$ export MY_VAR="Hello, Linux". The second command is \$ echo "\$MY_VAR", which outputs Hello, Linux. The third command is \$, followed by a cursor.

```
(kali@kali)-[~]  
$ export MY_VAR="Hello, Linux"  
  
(kali@kali)-[~]  
$ echo "$MY_VAR"  
Hello, Linux  
  
(kali@kali)-[~]  
$
```

GitHub Repo Link: [Harshit765G4/TudeDude-Assignments](https://github.com/Harshit765G4/TudeDude-Assignments)